

FPGA-Oriented Object Detection using Background Subtraction

RTL Design, Simulation and Verification

Project Team

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Project Guide

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Tools Used:

Verilog HDL

MATLAB

Simulink

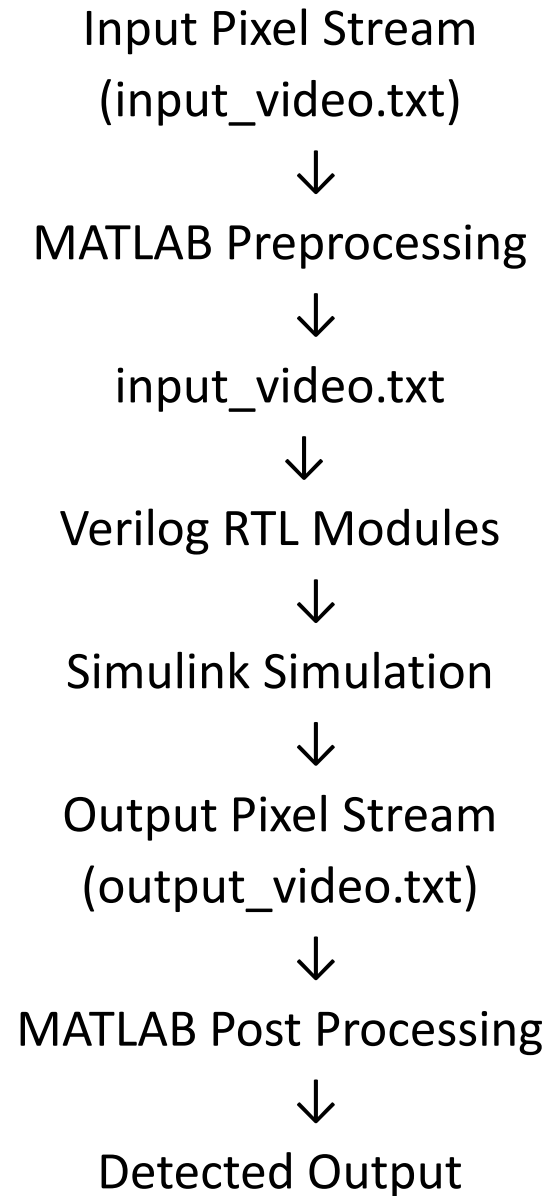
Xilinx ISE

Problem Statement

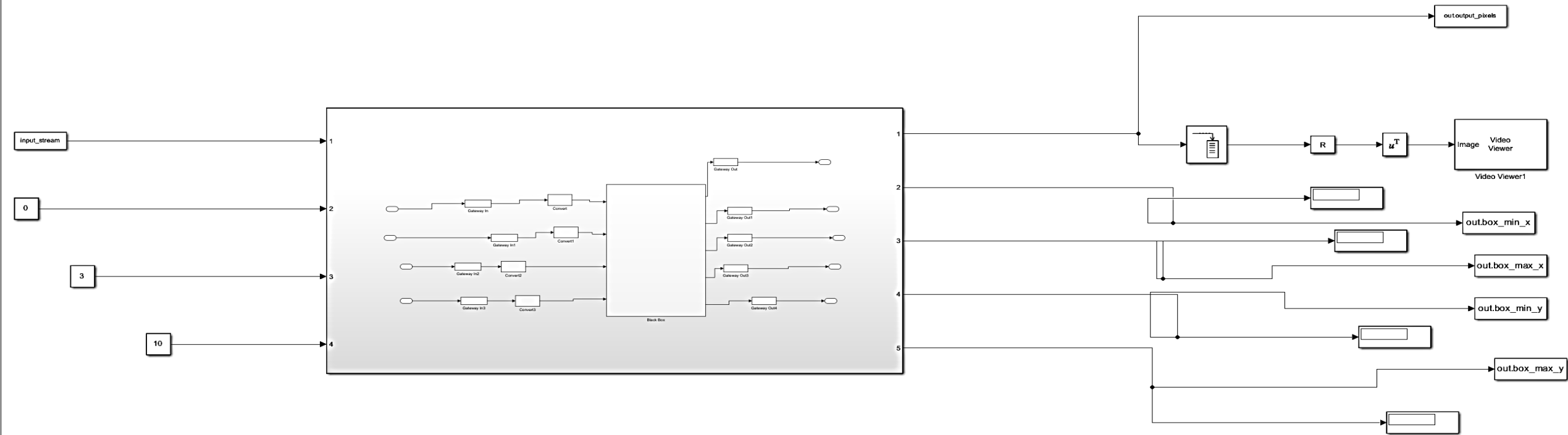
- Design and verify an RTL-based object detection pipeline capable of detecting moving objects using background subtraction.
- The objective was to understand how computer vision algorithms can be translated into RTL hardware modules and verified through simulation.



Workflow

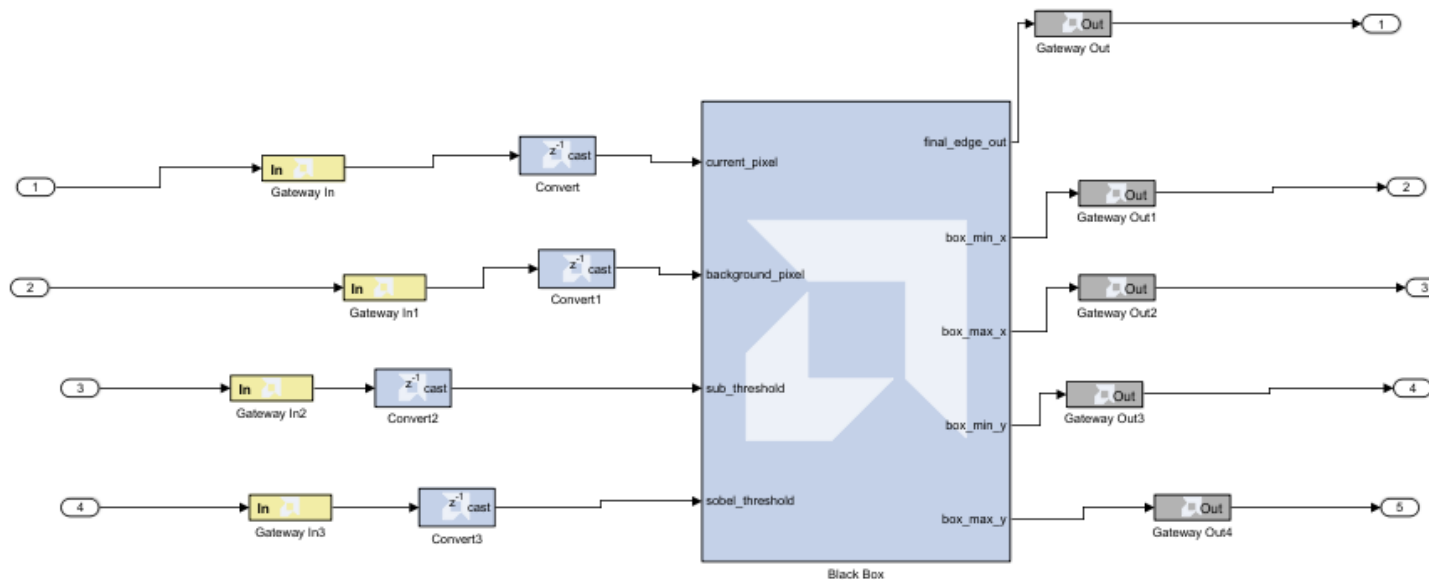


RTL Simulation Model



The Verilog RTL modules were integrated into Simulink for functional verification and output visualization.

RTL Processing Pipeline



RTL Modules

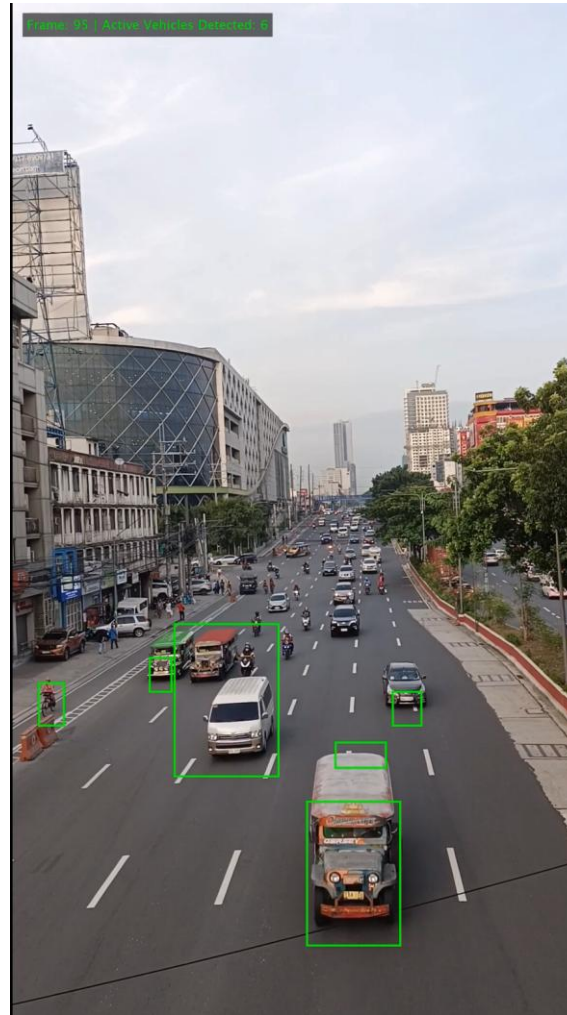
- Background Subtraction
- Line Buffer
- Sobel Edge Detection
- Bounding Box Generator
- Top-Level Tracking

Results

Input Frame



Output Frame



Simulation Outputs :

- Foreground detection
- Bounding box coordinates
- Processed pixel output

Technical Skills Demonstrated

- RTL Design
- Verilog HDL Development
- FPGA-Oriented Image Processing
- Functional Verification
- MATLAB/Simulink Integration
- Modular Hardware Design

Future Scope

- Hardware deployment on FPGA
- Performance optimization
- Resource utilization analysis